abstract-audit: AI-Native Academia figure and table sources

Main text

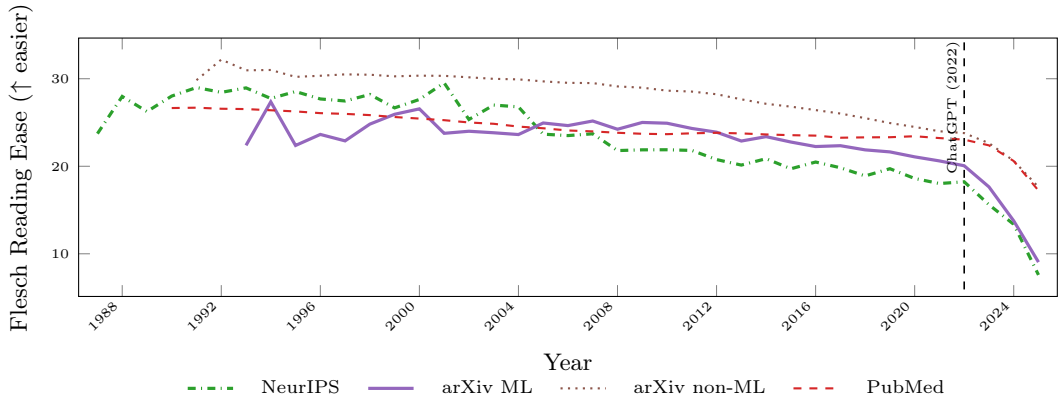


Figure 1: **Flesch Reading Ease on NeurIPS, arXiv ML, arXiv non-ML, and PubMed, 1987–2025.** Higher values indicate easier reading (↑). NeurIPS falls from approximately 24 in 1987 to approximately 8 in 2025. arXiv ML follows a similar trajectory. The arXiv non-ML baseline declines more slowly, indicating that the accelerated decline is specific to ML venues. Vertical dashed line marks the late 2022 availability of widely-used instruction-tuned writing assistants.

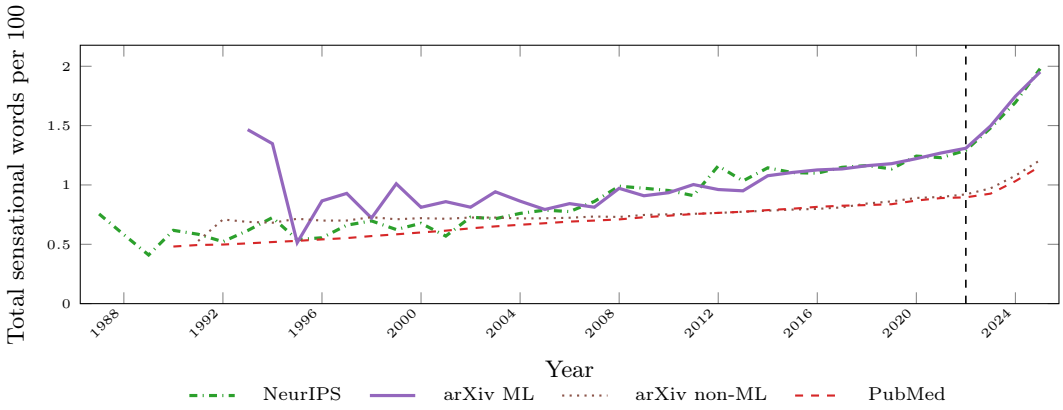


Figure 2: **Total sensational language per 100 abstract words, 1987–2025.** The NeurIPS total rate rises by approximately 80 percent between 2015 and 2025 (from 1.10 to 1.98 per 100 words). The arXiv non-ML baseline rises far more slowly. The divergence between ML and non-ML venues accelerates after the late 2022 availability of instruction-tuned writing assistants (vertical dashed line).

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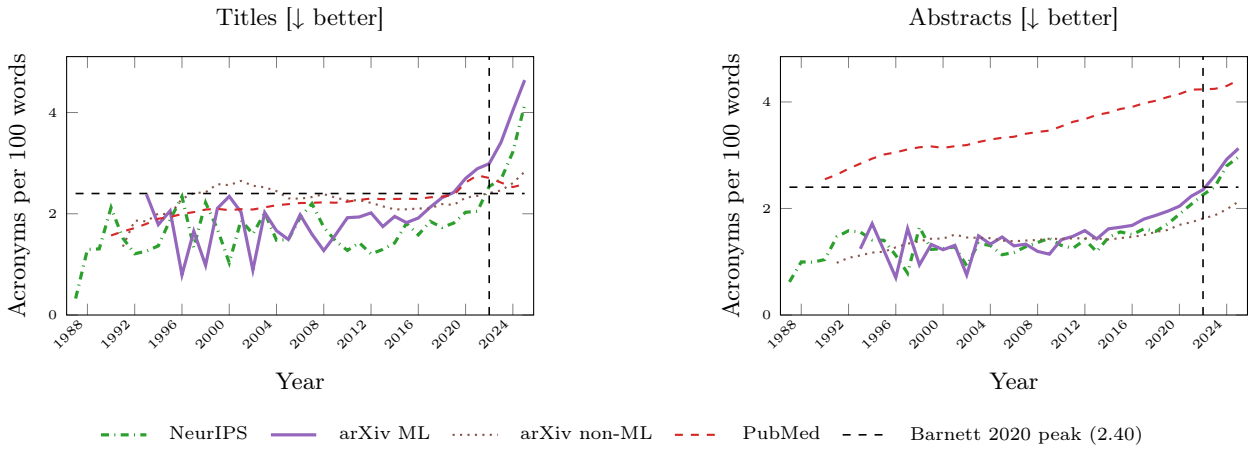


Figure 3: **Acronym density in titles (left) and abstracts (right), 1987 to 2025.** Lower values indicate fewer acronyms per word (↓ better). Dashed black horizontal line marks the Barnett and Doubleday science-wide peak of 2.40 per 100 words (2019). NeurIPS and arXiv ML diverge from the non-ML control around 2017 on titles and around 2020 on abstracts.

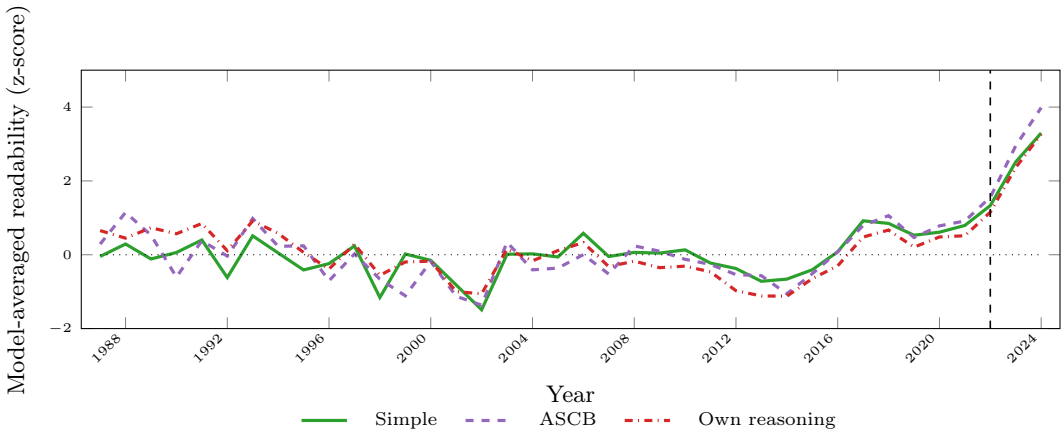


Figure 4: **LLM as judge readability on NeurIPS abstracts, 1987–2024.** Each curve is the mean z-score across all six open-weight models for one prompt variant, standardized to each model's own 1987–2022 baseline (dotted line at zero). All three prompts agree: LLM-judged readability is approximately flat before 2022 and rises sharply after the availability of instruction-tuned writing assistants (vertical dashed line), directly contradicting the classical readability decline in Figure 1.

Venue	Unique acronyms	1	2–9	10–99	100–999	1000+
NeurIPS	15,693	**31.2%**	58.3%	9.5%	1.0%	0.0%
arXiv ML	115,229	**28.9%**	59.1%	10.3%	1.5%	0.2%
arXiv non-ML	181,570	**32.0%**	50.7%	13.7%	3.1%	0.6%
PubMed	1,679,954	**31.1%**	48.4%	16.4%	3.2%	0.9%
B&D (2020)	1,112,345	30.0%	49.0%	—	—	—

Appendix

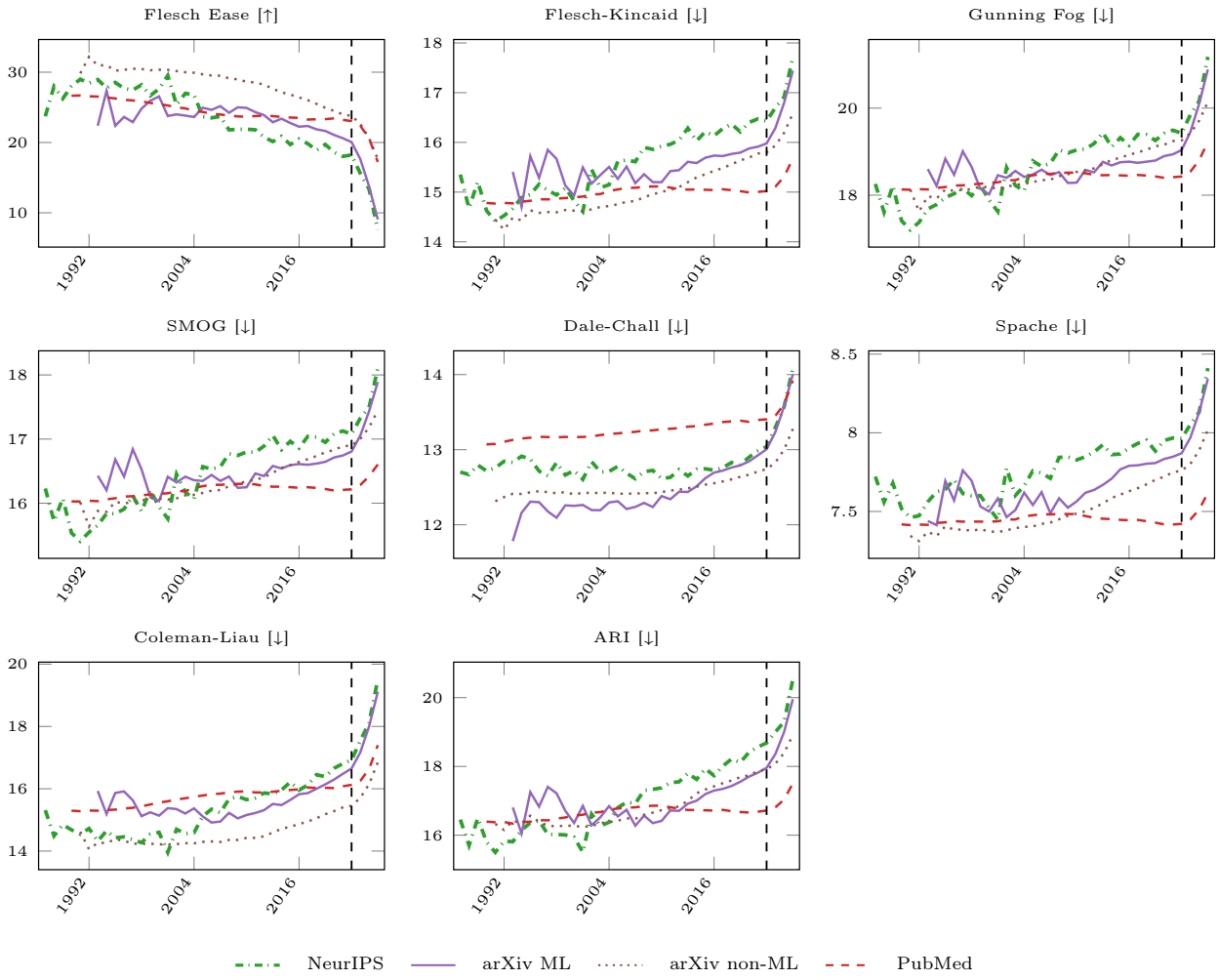


Figure 5: **Classical readability metrics, part A: Flesch family and grade-level formulas, 1987–2025.** Arrows indicate the direction of easier reading (↑ = higher is better; ↓ = lower is better). Every grade-level metric rises on NeurIPS and arXiv ML after 2015 and accelerates after 2020; Flesch Reading Ease drops correspondingly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part B with the remaining seven metrics appears in Figure 6.

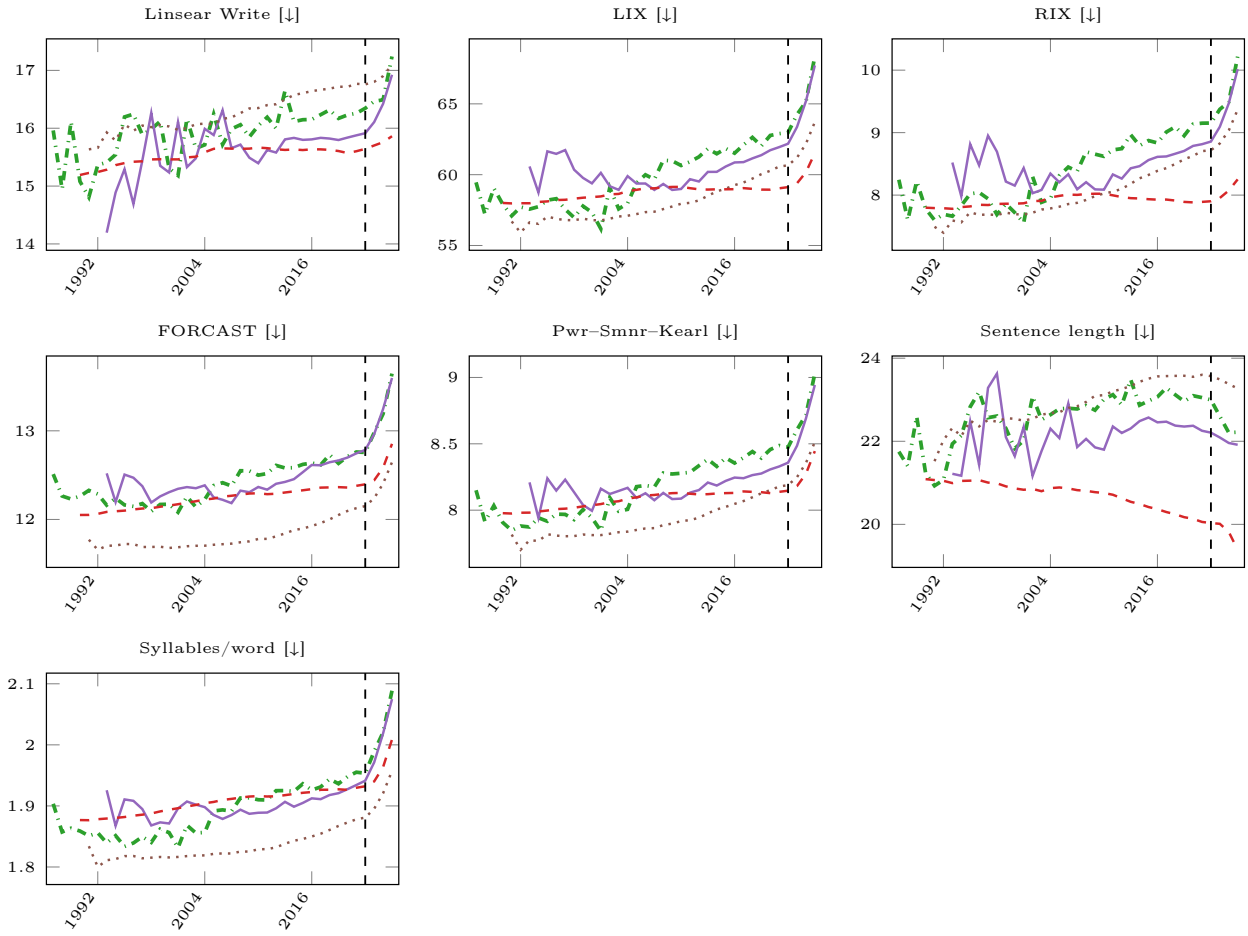


Figure 6: **Classical readability metrics, part B: Linsear Write, LIX, RIX, FORCAST, Powers-Sumner-Kearl, sentence length, and syllables per word, 1987–2025.** Arrows indicate the direction of easier reading (↓ = lower is better). Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part A with the Flesch family and the first eight grade-level formulas appears in Figure 5.

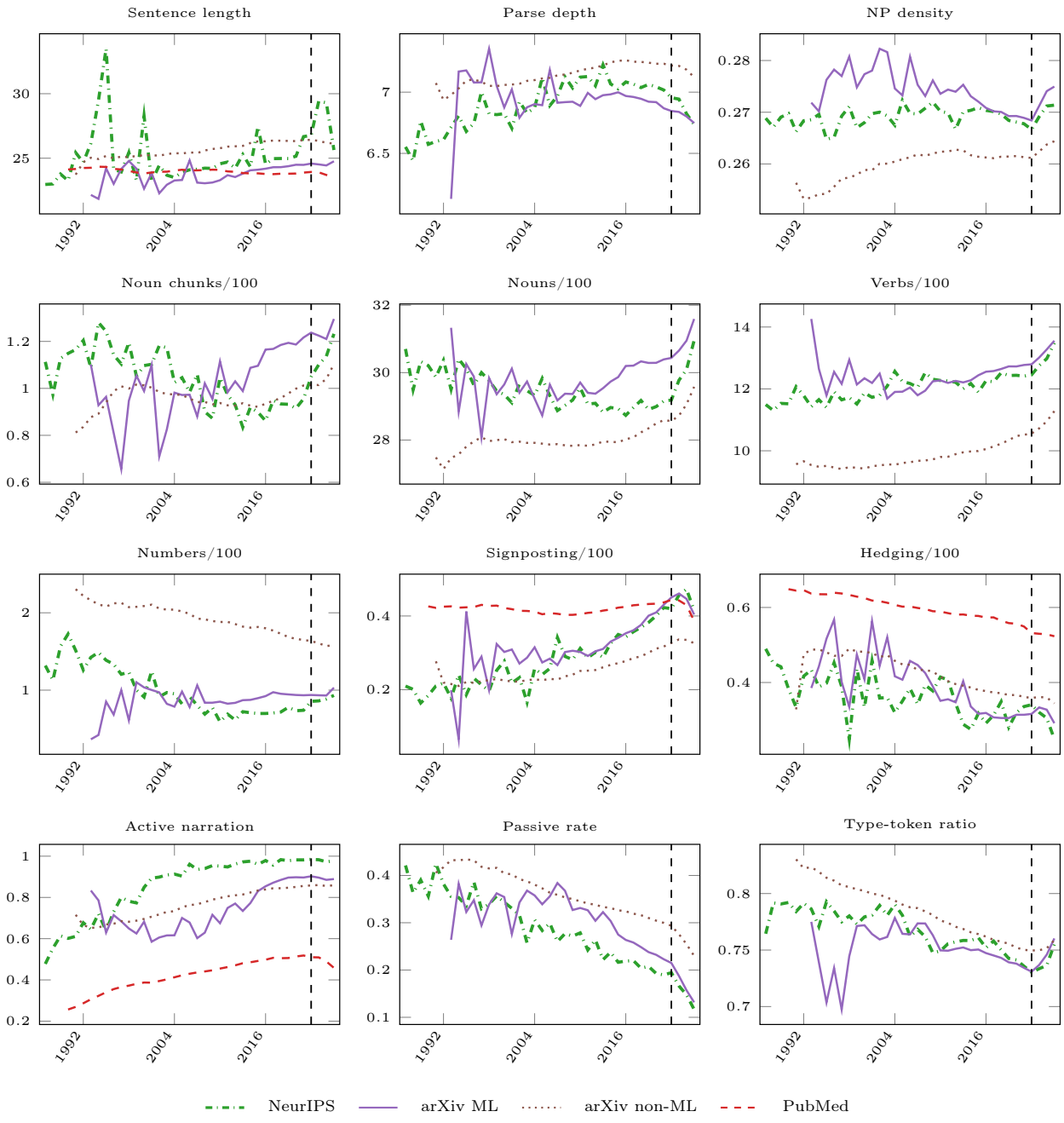


Figure 7: **Hohmann writing style metrics on NeurIPS, arXiv ML, arXiv non-ML, and PubMed, 1987–2025.** PubMed data are available only for the four metrics computed by lightweight regex and word-level rules (sentence length, signposting, hedging, and active narration). The remaining metrics depend on full spaCy dependency parsing and were not run on the 24.5 million paper PubMed corpus because of compute cost, so the corresponding panels show NeurIPS, arXiv ML, and arXiv non-ML only. Across these metrics, ML and non-ML venues are largely parallel. Only signposting, active narration, and passive rate show a clear widening of the ML versus non-ML gap after 2015.

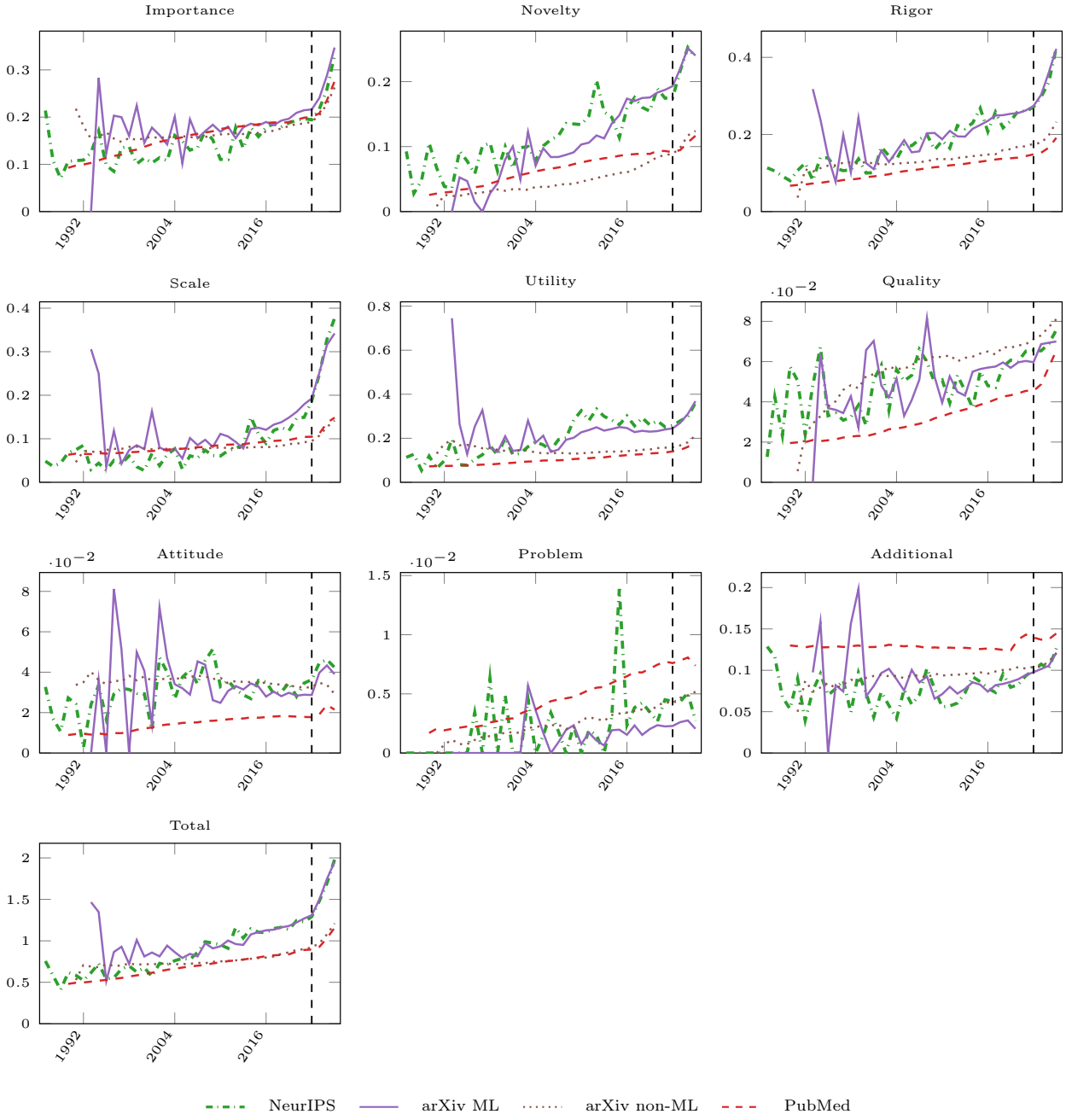


Figure 8: **All nine sensational language categories per 100 abstract words, 1987–2025.** Novelty and scale rise sharply after 2015, reaching roughly 2.1 times and 3.4 times their 2015 levels on NeurIPS by 2025 (1.6 and 2.7 times on arXiv ML); the total rate and rigor rise more modestly. Vertical dashed lines mark the late 2022 availability of instruction-tuned writing assistants.

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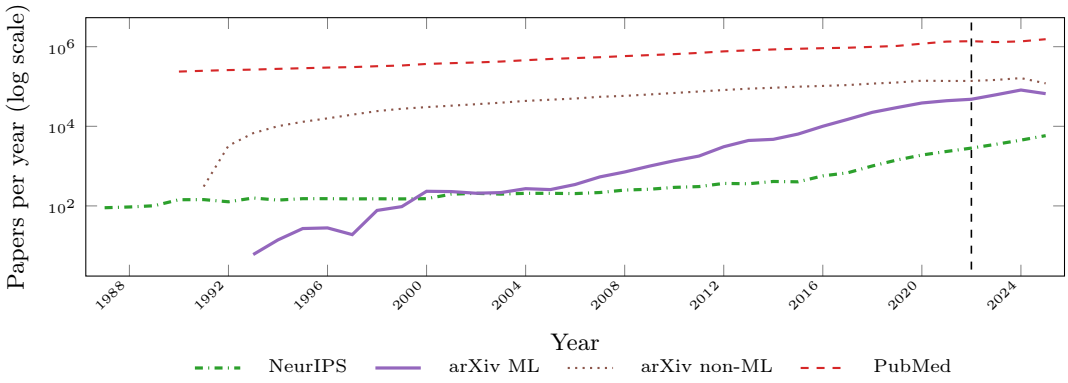


Figure 9: **Papers per year, 1987 to 2025.** arXiv ML submissions grew by roughly four orders of magnitude between 1993 and 2025. NeurIPS grew by a factor of roughly 65 between 1987 and 2025, from fewer than 100 accepted papers to approximately 5,800. PubMed and arXiv non-ML show sustained but slower growth over the same period. The volume increase is the structural backdrop against which the readability and acronym trends in Sections **??** and **??** must be interpreted.

Model	1987	2024	Δ	SD	Same
Gemma-3-27B	2.14	2.17	+0.03	0.02	98%
Gemma-4-31B	1.57	1.67	+0.10	0.03	95%
Llama-3.1-8B	1.97	2.20	+0.24	0.15	75%
Mistral-7B	2.62	3.04	+0.41	0.21	44%
Mixtral-8x7B	2.81	3.26	+0.44	0.09	83%
Qwen2.5-32B	2.18	2.10	-0.09	0.04	92%

Prompt	mean ρ	min	max	α
`simple`	0.396	0.286	0.487	0.786
`ascb`	0.370	0.241	0.478	0.754
`own_reasoning`	0.409	0.296	0.496	0.791

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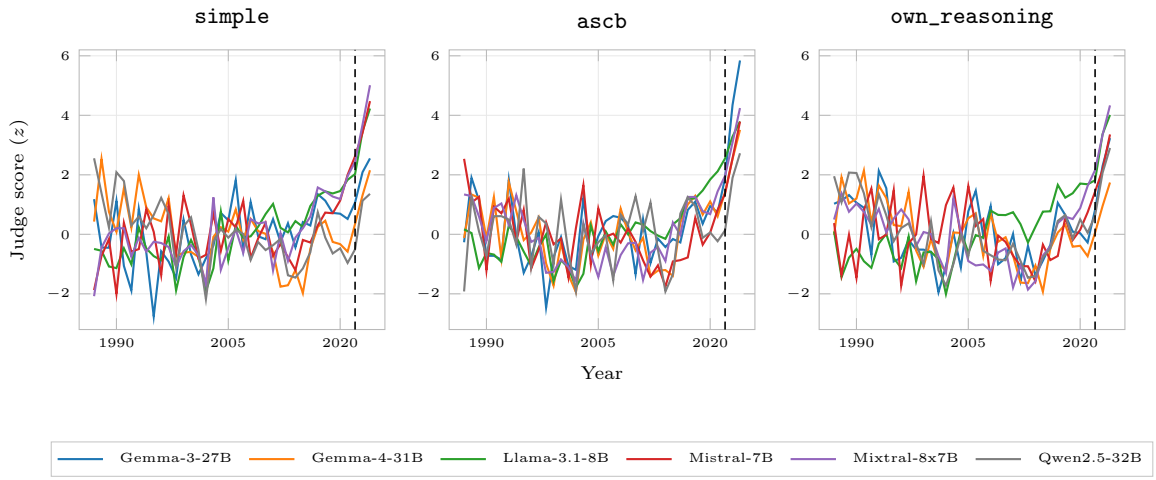


Figure 10: **Per-model judge scores by prompt template, standardized separately.** Each panel is one prompt template and each line one model, standardized to its own 1987 to 2022 mean and standard deviation. The dashed line marks late 2022. For each combination we fit a straight line through its yearly scores. The slope of that line is the average change in z per year. Before 2022 the slopes point in no consistent direction, nine of the 18 upward and nine downward, and none is steeper than 0.08 z per year, which is small against a year-to-year spread of about 4 z. After 2022 all 18 point the same way, rising by between 1.24 and 3.94 z. The model-averaged series in Figure 4 does not show the pre-2022 level differences or the threefold spread in the size of the rise, which follows from the differing baseline standard deviations.

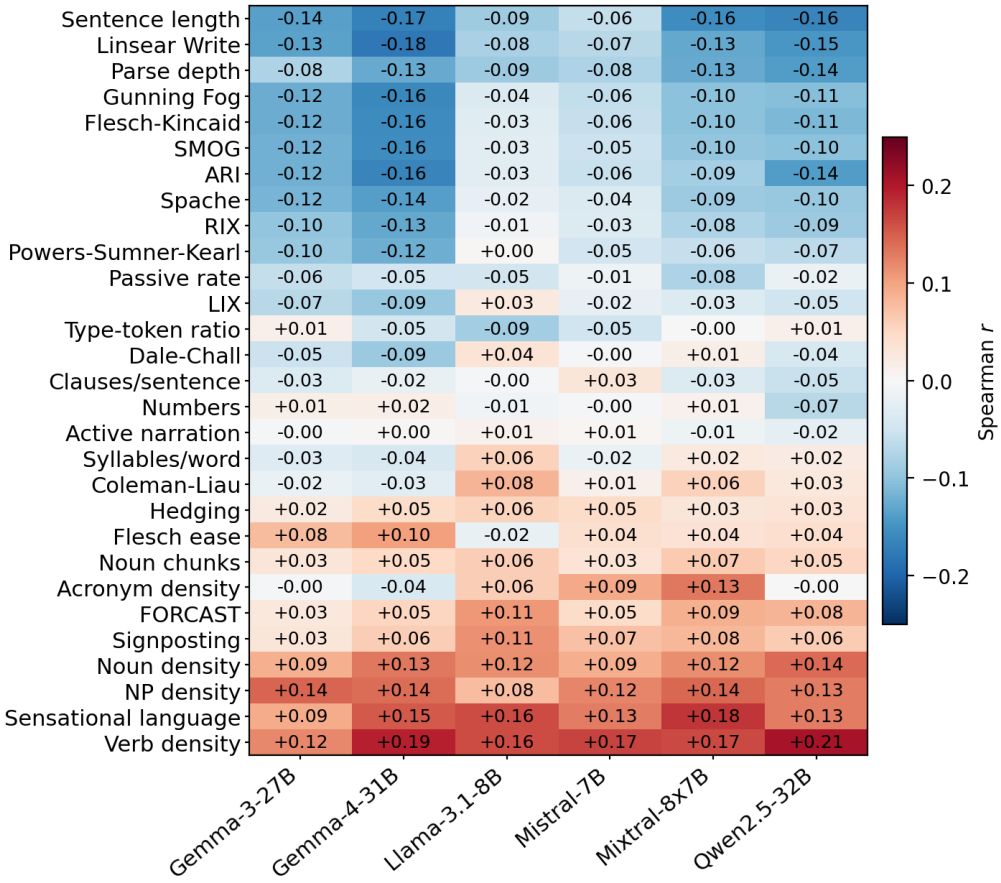


Figure 11: **What the LLM judge correlates with.** Per-paper Spearman correlation between each writing metric and each model's judge score, over 24,754 NeurIPS abstracts. A model's score for an abstract is the median of its nine generations, three prompts by three sampled runs. Rows are sorted by the correlation with the mean of those six per-model scores (1987–2024, the years the judge was scored). The judge score runs 1 to 5 with higher meaning easier, so each cell's sign says which way that feature moves the judge. **Red (positive): more of that feature is judged *more* readable**, as for verb density, sensational language and noun phrase density. **Blue (negative): more of it is judged *less* readable**, as for sentence length and syntactic complexity, which is the direction the classical formulas also take. Near-white cells, including acronym density, indicate near indifference. These are correlations, not causal claims, and every effect is small ($|r| < 0.25$).